Beyond Factual QA: Extending the Low-Resource Hallucination-Detector Robustness Phenomenon to Dialogue Generation and a Broader Set of Languages

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Dataset source and description

We use three datasets, one for each part of the study.

- **mTREx**: the base paper's own data (Datta et al., EACL 2026). Factual questions in English, German, Hindi, Bengali and Urdu, in three categories (capitals, country, official language). Each row has a source entity and one or more accepted answers. Raw files hold 2,374 to 10,000 rows. It is our reference setting.
- **X-FACTR** (Jiang et al., EMNLP 2020): 46 Wikidata relations, 46,000 facts, and relation templates in 23 languages. Entity labels and aliases exist for 21 non-English languages. We use it for breadth across resource levels.
- **PsiloQA**: 70,044 questions over 14 languages, each with a Wikipedia passage and an answer written by one of 26 language models, with the hallucinated spans marked. We use it for long answers with span labels.

Data collection method

One script, scripts/download_data.sh, fetches all three datasets, so anyone can rebuild them. mTREx comes from the authors' public repository (MIT licence, commit 644d655) and X-FACTR from its GitHub repository (commit c1938ec; the repository has no licence file and its data derives from Wikidata and Wikipedia). PsiloQA comes from Hugging Face (s-nlp/PsiloQA, CC BY 4.0). Raw data stays out of git. We commit the processed files and configuration, and log every decision that changed the data in data/README.md.

Preprocessing steps

We convert all three datasets to one record layout (uid, dataset, language, relation, prompt, gold, answer, spans, label, pair_uid, meta). Every random choice uses seed 42 and a stable

hash order.

- **mTREx**: split each target on the separator for multiple answers, then normalise (NFKC, whitespace, lower case). Drop empty rows and repeated sources, which takes 66,874 rows to 65,578. The smallest remaining cell sets one size for all, so every language and category keeps 2,140 rows, 32,100 records in total. Questions use the authors' own prompt file.
- **X-FACTR**: group languages by entity-label count (high at least 15,000; medium 4,000 to 14,999; low below 4,000) and choose languages whose templates need no gender or case morphology: high ja, zh, vi; medium tl, sw; low yo, ilo. We dropped Punjabi because its labels are in Shahmukhi while its templates are in Gurmukhi. No relation uses gendered wording in these languages, so all 46 relations stay. We keep a fact for a language only if its subject and object have labels in that language and in English. The prompt is the template with the subject filled in and a blank for the answer. Gold answers are the object label plus aliases. We order facts by how many chosen languages cover them, then by hash, and cap each relation at 30 facts. The result is 834 to 1,380 facts per language, 1,796 shared English facts and 9,977 records in total. Each non-English record points to its English twin.
- **PsiloQA**: pool the train, validation and test splits, keep English, Hindi, Chinese, Finnish and Arabic (39,869 rows), remove 13 repeated question-answer pairs (there were 0 empty answers), and set the label to 1 when the answer has at least one annotated span. Each language keeps 1,000 items, sampled in proportion to its own generator models, for 5,000 records in total. Generators differ by language, so an identical model mix across languages is impossible.

Brief EDA (key insights)

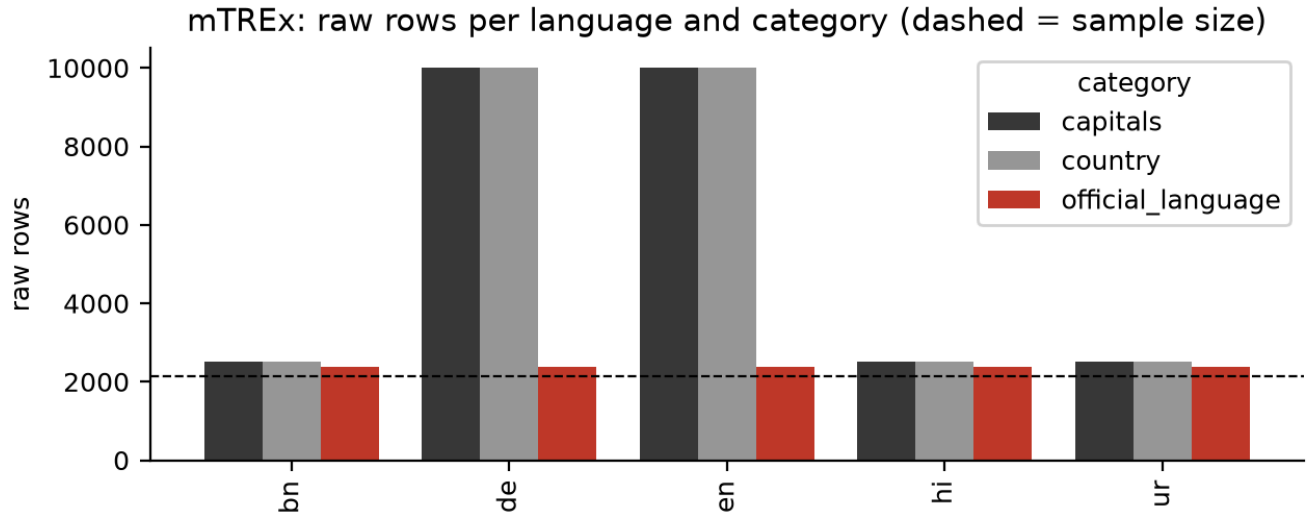


Figure 1. mTREx raw rows per language and category. The dashed line is the common sample size.

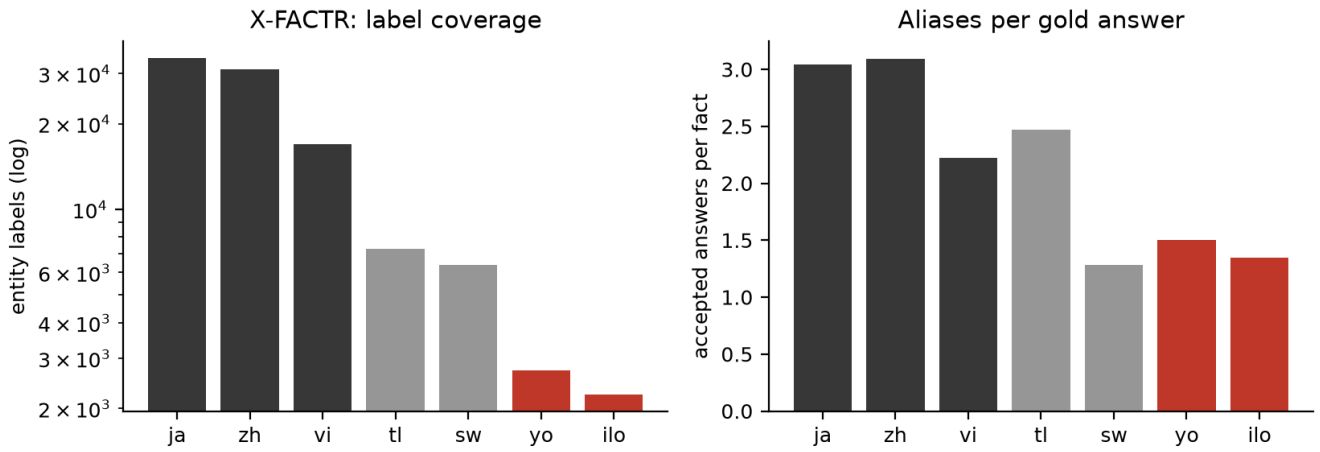


Figure 2. X-FACTR entity labels per chosen language (left, log scale) and accepted answers per fact (right). Dark: high resource, grey: medium, red: low.

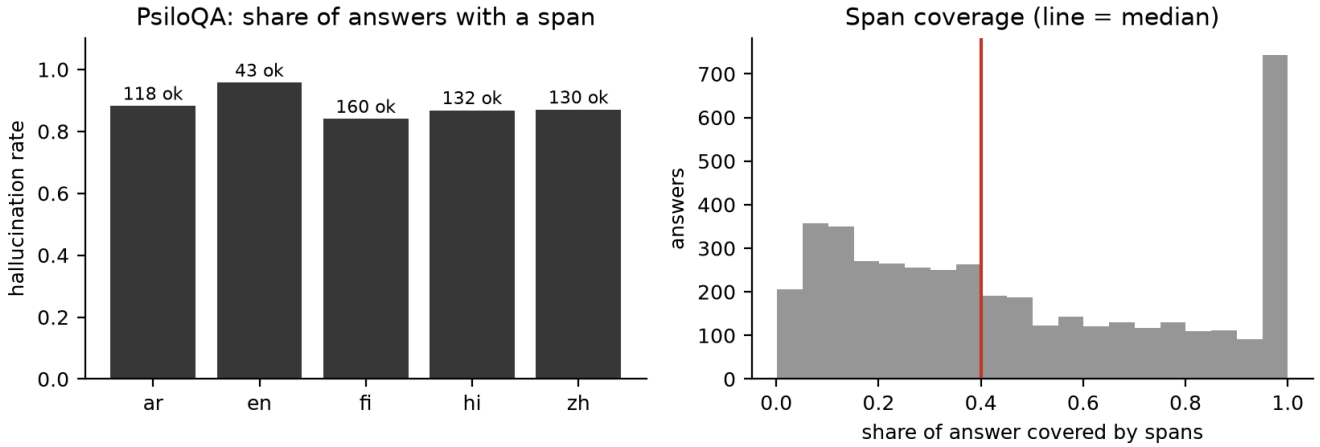


Figure 3. PsiloQA hallucination rate with the number of correct answers per 1,000 (left) and the share of each hallucinated answer covered by spans (right).

- The smallest mTREx file (Hindi, official language) limits the sample for all languages. About 29% of mTREx questions accept more than one answer in every language.
- X-FACTR label coverage falls from 33,954 (ja) to 2,241 (ilo), and accepted answers per fact fall from 3.09 to 1.28. Fewer accepted spellings make the substring matcher stricter in low-resource languages, so Phase 3 checks 50 answers per language by hand.
- PsiloQA answers are mostly hallucinated (rate per language: ar 0.88, en 0.96, fi 0.84, hi 0.87, zh 0.87). English has only 43 correct answers in 1,000, and no language has more than 160, so AUROC on PsiloQA will be noisy and class balance needs a decision before Phase 3.
- Spans are long: the median hallucinated answer is 40% covered by spans, with 1.5 spans on average. Generator models differ by language, so language and generator effects overlap and the bias check will report results per generator.

Final dataset summary

Dataset	Languages	Rows	Label	Intended use
mTREx	bn, de, en, hi, ur	32,100	gold match, assigned in Phase 3	reference setting
X-FACTR	en, ilo, ja, sw, tl, vi, yo, zh	9,977	gold match, assigned in Phase 3	breadth across resource levels
PsiloQA	ar, en, fi, hi, zh	5,000	1 if the answer has a span	long answers, span labels

Code, processed data and notebook (tag phase2): https://github.com/ahmad1khan2/f26-g6-hallucination-detectors